

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

Advanced Web Development and Frameworks (ITUE301)

Practical File – ODD Semester 2025–26

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Practical 1:

Introduction to React and Component Architecture

Problem Statement:

- Create a React application using Vite for a student portfolio page.
- The application must include at least 4 reusable components: Header, About, Skills, and Footer.
- Each component must be independently structured and composed into a single-page layout.
- No logic or JSX should be duplicated across components.
- Props must be used to pass at least one piece of data into a minimum of 2 components.

Solution:

HEADER –

```
function Header({ title }) {  
  return (
```

```
    <header>
      <h1>{title}</h1>
    </header>
  );
}
export default Header;
```

ABOUT :

```
function About() {
  return (
    <section>
      <p>Hi there! This is my student portfolio.</p>
    </section>
  );
}
export default About;
```

SKILLS :

```
function Skills({ skillList }) {
  return (
    <ul>
      {skillList.map((s) => (
        <li key={s}>{s}</li>
      ))}
    </ul>
  );
}

export default Skills;
```

FOOTER :

```
function Footer() {
  return (
    <footer>
      <p>Student Portfolio @2026</p>

    </footer>
```

```
);  
}
```

```
export default Footer;
```

Output :

Student Portfolio

Hi there! This is my student portfolio.

- HTML
- CSS
- JavaScript
- React
- Python

Student Portfolio @2026

KEY QUESTIONS :

- What is the role of each component in the overall UI structure?

Ans: Header – Displays the website title or logo.

NavBar – Provides navigation links to different pages.

Home – Displays the main content of the home page.

Projects – Shows the list of projects.

Contact – Displays contact information or a contact form.

Footer – Displays copyright and other footer information.

- How does React re-render when props change?

Ans: When a parent component passes new **props** to a child component, React compares the new props with the previous ones. If they are different, React re-renders the child component and updates only the changed parts of the UI using the Virtual DOM. This makes updates fast and efficient.

- Why is component reusability important in large-scale applications?

Ans: Component reusability is important because it:

Reduces duplicate code.

Makes the application easier to maintain.

- Saves development time.
- Keeps the code organized and readable.
- Makes testing and debugging easier.
- Ensures a consistent user interface across the application.

Supplementary Problems :

- Add a theme color prop to the Header and apply it as an inline style.

Output:



- Add a NavBar component that highlights the active section.

Output:

